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POWER CONVERTER CONTROL LOOP GAIN ADAPTATION

Abstract

Described embodiments include a circuit having a comparator with first and second comparator inputs and a comparator output. The second comparator input is coupled to a ripple reference voltage terminal. A switch is coupled between the first comparator input and a ground terminal, and has a switch control terminal that is coupled to the comparator output. A capacitor is coupled between the first comparator input and the ground terminal. A first current source is coupled between a supply terminal and the first comparator input, and has a first current control terminal. A second current source is coupled between a current output terminal and the ground terminal, and has a second current control terminal.

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Background/Summary

BACKGROUND

[0001] This description relates to switching power converters. Switching power converters receive a direct-current (DC) input voltage and provide an output at a specified DC voltage within a specified current limit. A typical switching converter includes a power stage with switches, an output capacitor, and an inductor. A typical switching converter may also include a converter control circuit to control the switches of the power stage. The power converter may include the capability to transition from a first operational mode to a second operational mode in response to load changes (e.g. light load to heavy load, or heavy load to light load). However, these mode transitions can sometimes be problematic by causing inefficient operation and ripple on the output voltage and current.

[0002] Some switching power converters may operate in both a pulse width modulation (PWM) mode and in a pulse frequency modulation (PFM). In some current mode power converters, a timer-based PFM mode may be used to adjust the PFM frequency. The gain of the control loop while operating in PFM mode usually depends on the PFM pulse size and the gain of the PFM timer. However, the PFM pulse size can vary depending on the duty cycle ratio and the inductance of an output inductor while operating in PWM mode

SUMMARY

[0003] In a first example, a circuit includes a comparator having first and second comparator inputs and a comparator output. The second comparator input is coupled to a ripple reference voltage terminal. A switch is coupled between the first comparator input and a ground terminal, and has a switch control terminal that is coupled to the comparator output.

[0004] A capacitor is coupled between the first comparator input and the ground terminal. A first current source is coupled between a supply terminal and the first comparator input, and has a first current control terminal. A second current source is coupled between a current output terminal and the ground terminal, and has a second current control terminal.

[0005] In a second example, a control circuit includes a clock circuit having a clock output, and a logic circuit having first, second, third and fourth logic circuit inputs and first, second and third logic circuit outputs. The first logic circuit input is coupled to the clock output. A digital-to-analog converter (DAC) has a DAC input and a DAC output. The DAC input is coupled to the first logic circuit output. A voltage-to-current converter (V2I) having first and second V2I inputs and first, second and third V2I outputs, wherein the first V2I input is coupled to the DAC output.

[0006] A timer circuit has first, second, and third timer outputs, wherein the first timer output is coupled to the first V2I input, the second timer output is coupled to the second V2I input, and the third timer output is coupled to the third logic circuit input. An amplifier having first and second amplifier inputs and an amplifier output, wherein the first amplifier input is coupled to a voltage feedback terminal, the second amplifier input is coupled to a voltage reference terminal, and the amplifier output is coupled to the second V2I input.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 shows a block diagram for an example power converter system.

[0008] FIG. 2 shows a diagram of inductor current versus time for an example voltage converter system operating in pulse frequency mode.

[0009] FIG. 3 shows a block diagram for an example PFM timing circuit.

[0010] FIG. 4 shows a schematic diagram for an example PFM timing circuit with an adapted reference voltage.

[0011] FIG. 5 shows a block diagram for an example buck power converter having a PFM timing circuit with an adapted reference voltage.

[0012] FIG. 6 shows a block diagram for an example circuit of a buck control logic circuit.

DETAILED DESCRIPTION

[0013] In this description, the same reference numbers depict same or similar (by function and/or structure) features. The drawings are not necessarily drawn to scale.

[0014] Some switching power converters are designed to operate in both a pulse width modulation (PWM) mode and in a pulse frequency modulation (PFM), and to transitions from one mode to the other during operation. Usually, in valley current mode buck power converters, a control loop adapts an inductor current valley point during operation in PWM mode. When operating in PFM mode, the same control loop can be used to adapt the frequency of the PFM pulses by controlling a PFM timer. The PFM timer determines the frequency of the PFM pulses using the output voltage of the loop filter error amplifier.

[0015] The control loop gain depends on the PFM pulse size when the PFM timer is used to control the switching frequency in PFM mode. Variations in arise in the control loop gain due to variations in the inductance of the output inductor, duty cycle ratio, and PWM switching frequency. This variation in gain in PFM mode can cause degraded load transient performance, voltage disturbances, and mode bouncing during entry to and exit from PFM mode.

[0016] FIG. 1 is a block diagram for an example power converter system **100**. The power converter system **100** includes a power supply **102**, a power stage **106**, an output capacitor C.sub.OUT **144**, a load **142**, and a controller **148**. The power stage **106** includes a high-side (HS) switch **120**, a low side (LS) switch **128**, and an inductor **136** coupled to the HS switch **120** and the LS switch **128** at a switching terminal **135**. In at least some examples, the HS switch **120**, LS switch **128**, and related drivers are components of an integrated circuit (IC), and the inductor **136** is an external component separate from the IC. In at least one example, power converter system **100** is a buck power converter, which implies that the output voltage (V.sub.OUT) is lower than the input voltage (V.sub.IN).

[0017] The HS switch **120** is coupled between the input voltage terminal VIN **104** and the LS switch **128**, and has a control terminal that receives a control signal HS_CS **152**. Power supply **102** provides an input voltage V.sub.IN **104** to HS switch **120**. The LS switch **128** is coupled between the LS switch **128** and a ground terminal, and has a control terminal that receives a control signal LS_CS **154**. The HS switch **120** is coupled to the LS switch **128** at a switching terminal **135**, which has a voltage V.sub.SW. In at least one example, the HS switch **120** and the LS switch **128** are p-channel field-effect transistors (“PFETs”). The inductor **136** is coupled between the switching terminal **135** and an output voltage terminal **140** that provides an output voltage V.sub.OUT. Output capacitor C.sub.OUT **144** is coupled between the output voltage terminal **140** and the ground terminal. A load **142** is coupled between the output voltage terminal **140** and the ground terminal.

[0018] The controller **148** includes valley control circuitry **155**, peak control circuitry **158**, light load detection circuitry **164**, mode control logic **170**, and driver circuit **184**. Controller **148** has a first input coupled to the power supply **102** that receives the input voltage V.sub.IN **104**. Controller **148** has a second input coupled to the switching terminal **135** that receives the switching voltage V.sub.SW. Controller **148** has a third input coupled to the output voltage terminal **140** that receives the output voltage V.sub.OUT.

[0019] Valley control circuitry **155** has an input that receives a control signal CS1. In some examples, CS1 includes a valley threshold and an inductor current sense signal, and CS1 may be one or more signals. In at least one example, the valley threshold and the inductor current sense signal may be ramped. Peak control circuitry **158** receives a control signal CS2. In some examples, CS2 includes a peak threshold and an inductor current sense signal. Light load detection circuitry **164** receive control signal CS3. In at least one example, CS3 includes a control voltage and a light load threshold. In some examples, the control voltage controls the output current of the power stage **106** and indicates the output current level. Accordingly, one option to detect a light load condition involves monitoring the control voltage relative to a threshold.

[0020] Mode control logic **170** has a first input coupled to an output of the valley control circuitry **155**, a second input coupled to an output of the peak control circuitry **158**, and a third input coupled to an output of the light load detection circuitry **164**. Mode control logic **170** has a first output coupled to a first input of driver circuit **184**, and a second output coupled to a second input of driver circuit **184**. Mode control logic **170** is configured to provide a ripple control signal, Sel_i_ripple **178**, at a third output. Driver circuitry **184** has a first output HS_CS **152** which is coupled to the control terminal of HS switch **120**. Driver circuitry **184** has a second output LS_CS **154** that is coupled to the control terminal of LS switch **128**.

[0021] The controller **148** provides HS_CS **152** and LS_CS **154**, respectively, in response to the voltages V.sub.IN, V.sub.SW, V.sub.OUT and ground, and the operations of the valley control circuitry **155**, the peak control circuitry **158**, the light load detection circuitry **164**, the mode control logic **170**, and the driver circuit **184**. In some examples, the controller **148** may be configured to sense inductor current of the power stage and adjust an offset between a valley threshold and a peak threshold in response to a light load condition being detected by the light load detection circuitry **164**. The controller **148** may be configured to transition from a PWM mode to a PFM mode in response to the light load condition combined with a comparison indicating the sensed inductor current reaches the peak threshold. The comparison is performed, for example, by the peak control circuitry **158**.

[0022] The power stage **106** provides the output voltage V.sub.OUT at the output voltage terminal **140**, and provides V.sub.SW at the switching terminal **135** responsive to the voltages V.sub.IN **104**, HS_CS **152**, and LS_CS **154**. The HS switch **120** couples the input voltage terminal **104** to the switching terminal **135** responsive to HS_CS **152**, which increases the current in the inductor **136**. The LS switch **128** couples the ground terminal to the switching terminal **135** responsive to LS_CS **154**, which decreases the current in the inductor **136**. The load current provided to the load **142** is the average current through the inductor **136**.

[0023] The PFM pulse size may be selected to achieve good efficiency and to avoid high output voltage ripple or mode bouncing between the PFM and PWM modes. In some examples, the controller **148** is configured to optimize efficiency by entering the PFM mode at light loads when the inductor current falls below zero current while operating in the PWM mode. In the case of a light load, the maximum PFM output current is selected to be larger than the load current to avoid bouncing between PWM mode and PFM mode, and to avoid overshoots or undershoots in the output voltage V.sub.OUT **140**. The PFM pulse is selected to be larger than the current ripple in the PWM mode.

[0024] Conditions that may cause a variance in the inductor current ripple when operating in the PWM mode include variances in the input voltage V.sub.IN **104**, the output voltage V.sub.OUT **140**, the operating frequency, and the inductance of inductor **136**. In some cases, the inductance of the inductor **136** may vary up to $\pm 30\%$, and an example operating frequency may vary up to $\pm 20\%$. Also, in at least one example, the input voltage V.sub.IN **104** may vary between 2.5V and 40V. An example output voltage V.sub.OUT **140** may vary from 0.3 to 6V. Each of these variances can cause the inductor current ripple to vary. Due at least in part to the inductor current ripple variance, the appropriate peak current value for the PFM pulse in a high ripple condition can cause significant voltage ripple in a low ripple condition. Conversely, the appropriate PFM pulse for a low ripple condition can cause inefficiency and mode bounce in a high ripple condition.

[0025] The controller **148** adjusts the PFM pulse peak current value responsive to inductor current ripple in the PWM mode. In at least some examples, the peak control circuitry **158** controls the peak current limit in the PWM mode and defines the PFM pulse peak current value in the PFM mode. For example, the peak threshold may be a valley current level plus an offset. In some examples, the offset is digitally controlled. In the PFM mode, the controller **148** may set the valley current level to zero, and the reference level for the PFM peak current value is determined by the offset.

[0026] In the PWM mode, the controller **148** uses a fixed frequency valley current mode to regulate the output voltage $V_{sub.OUT}$ **140** by adapting the valley current level as needed. In the case of a current mode buck power converter operating in timer based PFM mode, a current control loop is used to adapt the PFM frequency. The PFM pulse is adapted based on the inductor current ripple in PWM mode to optimize the output voltage ripple and converter efficiency.

[0027] FIG. **2** shows a diagram **200** of inductor current versus time for an example voltage converter system operating in pulse frequency mode (PFM). Graph **210** shows a peak output current level for the voltage converter, I_{peak} . Graph **220** shows an average output current level for the voltage converter, I_{out} . Graph **202** shows the inductor current being delivered to the output voltage terminal versus time. Graph **230** shows the Start_pfm_pulse **230**, which triggers the start of each PFM cycle, versus time.

[0028] A pulse occurs on the Start_pfm_pulse signal **230** at time t_0 . In response to the pulse, the high side switch is turned on coupling the input voltage from the power supply to the switching terminal. This causes current to flow through the inductor and to the output capacitance creating a ramp in the inductor current **202**. The inductor current **202** continues to rise until it reaches the peak output current level I_{peak} **210**. In response to the inductor current **202** reaching the peak output current level I_{peak} **210**, the high side switch is turned off and the low side switch is turned on, coupling the switching terminal to the ground terminal. The inductor current begins decreasing and continues to decrease until it reaches zero. The converter will remain in a high impedance state and the inductor current **202** will remain at zero until the next pulse on the Start_pfm_pulse signal **230** occurs.

[0029] The time from the rising edge of the Start_pfm_pulse signal **230** until the inductor current **202** ramps back down to zero is T_{pulse} **212**. Subsequently, another pulse occurs on the Start_pfm_pulse signal **230**, and the inductor current **202** begins to ramp up again until it reaches the peak output current level I_{peak} **210**. The time from the rising edge of a particular Start_pfm_pulse signal **230** until the rising edge of the subsequent Start_pfm_pulse signal **230** is T_{SW} **216**, which is the PFM pulse period. The average output current I_{out} **220** is determined by the period (or frequency) and the magnitude of the PFM pulses. $I_{sub.OUT}$ **220** is the average inductor current, and its value can be calculated using equation (1) as half of the peak current multiplied by the ratio of the pulse width to the period between consecutive pulses.

[00001]
$$I_{out} = \frac{I_{peak}}{2} * \frac{T_{pulse}}{T_{SW}} \quad (1)$$

[0030] FIG. **3** shows a block diagram for an example PFM timing circuit **300**. Amplifier **306** has a first input coupled to a feedback voltage terminal that provides a feedback voltage V_{fb} **302**. The feedback voltage V_{fb} **302** is a voltage proportional to the output voltage of the power converter. Amplifier **306** has a second input coupled to a reference voltage terminal that provides a reference voltage V_{ref} **304**. The reference voltage V_{ref} **304** is proportional to the desired voltage at the output voltage terminal. Amplifier **306** has an output that is coupled to the control terminal of a current source gm_{pfm} **314**. Resistor **310** and capacitor **312** are coupled in series between the output of amplifier **306** and a ground terminal.

[0031] Current source gm_{pfm} **314** has a first terminal coupled to a voltage source, a second terminal coupled to a switch **318** and capacitor C_{pfm} **322**, and a control terminal receiving the signal V_{ctrl} **308** from the output of amplifier **306**. Comparator **324** has a first input coupled to the second terminal of current source gm_{pfm} **314**, and receives a signal V_{timer_ramp} **320**. Comparator **324** has a second input coupled to the reference voltage terminal, and receives the reference voltage V_{ref} **304**. Capacitor **322** C_{pfm} is coupled between the first input of comparator **324** and the ground terminal. Comparator **324** provides a signal Start_pfm_pulse **326** at its output.

[0032] The circuitry of PFM timing circuit **300** converts a compensating voltage, V_{ctrl} **308**, at the output of amplifier **306** to a time period. This conversion is accomplished using compensating voltage V_{ctrl} **308** to control the amount of current provided by current source gm_{pfm} **314** to

capacitor C_{pfm} 322, which creates a voltage V_{timer_ramp} 320 across capacitor C_{pfm} 322. The voltage V_{timer_ramp} 320 is provided to comparator 324, which compares that voltage to the reference voltage V_{ref} 304. The comparison of V_{timer_ramp} 320 to V_{ref} 304 controls the length of the pulse period t_{sw} 216. When the pulse period t_{sw} elapses, a subsequent PFM pulse is triggered by the Start_pfm_pulse 326.

[0033] Resistor 310 and capacitor 312 are coupled in series to the output of amplifier 306 and form a proportional-integral (P-I) compensator for the voltage V_{ctrl} 308. That voltage controls the transconductance (g_{sub.m}) of the timer, which operates similarly to a voltage-controlled oscillator in which the voltage V_{ctrl} 308 is controlling the oscillation frequency.

[0034] The Start_pfm_pulse signal 326 is provided by the output of comparator 324.

Start_pfm_pulse signal 326 is provided at the control terminal of switch 318 to control the switch. In response to the Start_pfm_pulse signal 326 causing switch 318 to open, the current from current source gm_{pfm} 314 begins charging capacitor C_{pfm} 322, causing the voltage V_{timer_ramp} 320 to increase linearly. In response to the voltage V_{timer_ramp} 320 crossing the reference voltage V_{ref} 304, the output of comparator 324 goes high, triggering a Start_pfm_pulse 326.

[0035] The buck controller generates the Start_pfm_pulse 326, and switch 318 is opened. The power stage of the voltage converter generates a current pulse through the output inductor, and the voltage timer ramp cycle begins again with a subsequent rising Start_pfm_pulse 326. As the voltage at V_{ctrl} 308 increases, the current provided by current source gm_{pfm} 314 increases, and the voltage ramp of V_{timer_ramp} 320 across C_{pfm} 322 becomes steeper (i.e. higher slew rate). As the voltage ramp of V_{timer_ramp} 320 becomes steeper, the Start_pfm_pulse 326 triggers earlier, which generates faster pulses and increases the average output current I_{sub.OUT} 220.

[0036] If the output current I_{sub.OUT} 220 demand increases, the output voltage may begin to drop. A drop in the output voltage causes a drop in the feedback voltage V_{fb} 302, which increases the voltage V_{ctrl} 308 at the output of amplifier 306. Increasing the voltage V_{ctrl} 308 increases the amount of current supplied by current source gm_{pfm} 314, charging the capacitor C_{pfm} 322 more quickly, which makes the output of comparator 324 trip faster. As a result, the period in which the inductor current is zero decreases and a new Start_pfm_pulse 326 is generated sooner, which increases the average output current I_{sub.OUT} 220.

[0037] The PFM gain is the ratio of the change in average output current I_{sub.OUT} to the change in voltage V_{ctrl} 308 at the output of amplifier 306. There is a voltage-to-current conversion from the voltage V_{ctrl} 308 to the current output from current source gm_{pfm} 414, and this circuitry forms a voltage-to-current converter. An equation for calculating the PFM gain is given by equation (2):

$$[00002] \text{Gain}_{\text{pfm}} = \frac{I_{\text{out}}}{V_{\text{ctrl}}} = \frac{T_{\text{pulse}} * g_{\text{m}_{\text{pfm}}}}{V_{\text{ref}} * C_{\text{pfm}}} \quad (2)$$

The PFM gain depends on the pulse width, the transconductance of the current source, the capacitance of C_{pfm} 322 and the reference voltage V_{ref} 304. Different applications may have different input voltages and output voltages, and different tolerances for the inductance of the inductor.

[0038] These variations in parameters can lead to higher current ripple in some applications, or smaller current ripple in other applications. The magnitude of the current ripple affects the PFM gain, causing the gain from the voltage V_{ctrl} 308 at the output of amplifier 306 to output current to vary significantly. This variation in gain can cause significant problems including increased ripple on the output current and voltage, control loop instability, and unstable transitions between PWM mode and PFM mode.

[0039] A potential solution to the problems caused by the variation of PFM gain is to change the pulse frequency based on the inductor current ripple. The reference voltage level provided to the PFM timer comparator can be adapted and varied in response to the PFM pulse amplitude. This makes the reference voltage V_{ref} no longer a constant DC level voltage, but instead adjustable.

[0040] FIG. 4 shows a schematic diagram for an example PFM timing circuit with an adapted reference voltage. The signal V_{ctrl} 408 is a signal proportional to the difference in voltage between the output voltage of a regulator and a reference voltage representing a desired output voltage. In at least one case, the signal V_{ctrl} 408 is provided from the output of a voltage error amplifier. The signal V_{ctrl} 408 is provided to the control terminal of a current source gm_{pfm} 414.

[0041] The current source gm_{pfm} 414 has a first terminal coupled to a voltage source, a second terminal coupled to a switch 418, and a control terminal receiving the signal V_{ctrl} 408. Comparator 424 has a first input coupled to the second terminal of current source gm_{pfm} 414 that receives a signal V_{timer_ramp} 420. Comparator 424 has a second input that receives a reference voltage V_{ripple} 436.

[0042] Capacitor 422 C_{pfm} is coupled between the first input of comparator 424 and the ground terminal. Comparator 424 provides a signal $Start_{pfm_pulse}$ 426 at its output, which is coupled to the control terminal of switch 418. Switch 418 is coupled between current source gm_{pfm} 414 and the ground terminal, and has a control terminal coupled to the output of comparator 424. Capacitor C_{pfm} 422 is coupled between the first input of comparator 424 and the ground terminal.

[0043] Digital-to-analog converter (DAC) 434 has an input that receives a digital signal Sel_i_ripple 432. In one example, the digital signal Sel_i_ripple 432 comprises six bits, but Sel_i_ripple 432 could comprise more or less than six bits in other examples. The output of DAC 434 is coupled to the second input of comparator 424 and to a control terminal of current source 438. DAC 434 provides a signal V_{ripple} 436 at its output that is an analog conversion of Sel_i_ripple 432. The signal V_{ripple} 436 is also provided to the control terminal of current source 438. Current source 438 provides a current signal I_{ripple} 440.

[0044] The magnitude of V_{CTRL} 408 controls the magnitude of the current that is provided by current source gm_{pfm} 414 for charging capacitor C_{pfm} 422. The output of DAC 434, which is the signal V_{ripple} 436, provides the reference voltage that the signal V_{timer_ramp} 420 is compared to using comparator 424. The signal V_{ripple} 436 also controls the reference current level for an output current peak comparator. The reference current level is set by controlling the magnitude of the current provided by current source 438 as the signal I_{Ripple} 440. The reference current level for I_{peak} is equal to the sum of I_{ripple} and I_{CTRL} . The value of I_{CTRL} defines the valley current level, and the value of I_{ripple} defines the current ripple between comparators.

[0045] The signal V_{ripple} 436, which is the output of DAC 434, determines the peak current of the PFM pulse, and provides an input to comparator 424. At its output, comparator 424 provides the $Start_PFM_pulse$ that triggers each respective PFM pulse. The current ripple in PWM mode is measured, and the difference between the peak value and the valley value is used to generate a reference voltage for comparator 424, which allows the PFM gain to remain constant under different operating conditions. The control loop gain can be designed to be the same in PFM mode as it is in PWM mode.

[0046] FIG. 5 shows a block diagram for an example buck power converter 500 having a PFM timing circuit with an adapted reference voltage. The switch clock generator 502 provides a clock signal Clk_SW , which is a first input to buck control logic circuit 508. A second input to buck control logic circuit 508 is $Valley_Comp$ 504, which is the output signal of a current comparator that detects a zero current level in PFM mode. In PWM mode, a valley comparator defines the controlled valley current level. A third input to buck control logic circuit 508 is $Peak_Comp$ 506, which is the output of a current comparator that detects a peak inductor current level. A fourth input to buck control logic circuit 508 is the signal $Start_{pfm_pulse}$ 426. Buck control logic circuit 508 may include digital logic gates or can be a processor or a microcontroller.

[0047] A first output of buck control logic circuit 508 is $Buck_pwm$ 510, which is provided as a first input to driver circuit 184. A second output of buck control logic circuit 508 is $Buck_hiz$, which is provided as a second input to driver circuit 184. A first output of driver circuit 184,

HS_CS **152**, is coupled to the control terminal of HS switch **120**. A second output of driver circuit **184**, LS_CS **154**, is coupled to the control terminal of LS switch **128**. HS switch **120** is coupled to LS switch **128** at the switching terminal SW **135**. Inductor **136** is coupled between the switching terminal SW **135** and the output voltage terminal VOUT **140**. Output capacitor COUT **144** is coupled between the output voltage terminal VOUT **140** and the ground terminal.

[0048] FIG. **6** shows a block diagram for an example circuit **600** for buck control logic circuit **508**. Buck control logic circuit **508** includes switch control circuitry **616**, state machine circuitry **610**, and ripple control circuitry **618**. The switch clock signal Clk_SW is a first input to switch control circuitry **616**. A second input to switch control circuitry **616** is Valley_Comp **504**, and a third input to switch control circuitry **616** is Peak_Comp **506**. A fourth input to switch control circuitry **616** is the signal Start_pfm_pulse **426**.

[0049] A first input of state machine circuitry **610** is coupled to the third input of switch control circuitry **616**, and receives the Peak_Comp **506** signal. A second input of state machine circuitry **610** receives the signal PFM_Entry_Comp **608**. A first output of the state machine circuitry **610** provides the signal PFM_Peak_Detection **614** to ripple control circuitry **618**. A second output of the state machine circuitry **610** provides the signal PFM_Mode **612**, which is provided as a second input to ripple control circuitry **618**, and as a fifth input to switch control circuitry **616**.

[0050] The state machine circuitry **610** is configured to receive Peak_Comp **506** and PFM_Entry_Comp **608**, and provide a PFM mode signal PFM_Mode **612**, and a PFM peak detection signal PFM_Peak_Detection **614**. The ripple control circuitry **618** is configured to receive PFM_Peak_Detection **614** and PFM_Mode **612**, and provide the ripple control signal Sel_I_Ripple **432**. After PFM_Peak_Detection **614** is asserted, the ripple control circuitry **618** decreases the ripple control signal Sel_I_Ripple **432** periodically until the signal PFM_Mode **612** is asserted or PFM mode entry is canceled. The switch control circuitry **616** is configured to receive CLK_SW, Valley_Comp **504**, Start_PFM_pulse **426**, Peak_Comp **506** and PFM_Mode **612**, and to provide Buck_pwm **510** and Buck_hiz **520**.

[0051] The input of DAC **434** is coupled to buck control logic circuit **508** and receives the ripple control signal Sel_I_Ripple **432**. In at least one example, the ripple control signal Sel_I_Ripple **432** is a six-bit digital word, but Sel_I_Ripple **432** can be more or less bits in other examples. The output of DAC **434** is coupled to a first input of voltage-to-current (V2I) circuit **524**.

[0052] Amplifier **306** has a first input coupled to a feedback voltage terminal that provides a feedback voltage V_fb **302**. The feedback voltage V_fb **302** is a voltage proportional to the output voltage VOUT **140** of the power converter. Amplifier **306** has a second input coupled to a reference voltage terminal that provides a reference voltage V_ref **304**. The reference voltage V_ref **304** is proportional to the desired voltage at the output voltage terminal VOUT **140**. Amplifier **306** has an output that is coupled to a second input of V2I circuit **524**. Resistor **310** and capacitor **312** are coupled in series between the output of amplifier **306** and the ground terminal. PFM timer circuit **526** has first, second and third outputs. The first and second outputs are coupled to the first and second inputs of V2I circuit **524**, respectively, and the third output provides the Start_pfm_pulse signal **426**.

[0053] Replica switches **530** include LS replica switches **532** and HS replica switches **534**. A first output of V2I circuit **524** is coupled to a first input of LS replica switches **532** and to a first input of HS replica switches **534**, and provides the control signal I_CTRL **518**. A second output of V2I circuit **524** is coupled to a second input of LS replica switches **532** and provides the signal I_OFFSET **526**. A third output of V2I circuit **524** is coupled to a second input of HS replica switches **534** and provides the signal I_RIPPLE **528**.

[0054] HS replica switches **534** generates differential sense and reference signals HS_sw_sense **540** and HS_vdd_sense **542** based on the current through HS switch **120** and control signals I_CTRL **518** and I_RIPPLE **528**. Control signals I_CTRL **518** and I_RIPPLE **528** generate voltage drops across HS replica switches **534** which is relative to the voltage drop across HS switch **120**.

LS replica switches **532** generates differential sense and reference signals LS_sw_sense **536** and LS_gnd_sense **538** based on the current through LS switch **128** and control signals I_CTRL **518** and I_OFFSET **526**. Control signals I_CTRL **518** and I_OFFSET **526** and ramp signals from artificial ramp circuit **522** generate voltage drops across LS replica switches **532** which is relative to the voltage drop across LS switch **128**.

[0055] Artificial ramp circuit **522** has a first output coupled to the first input of replica switches **530** and a second output coupled to the second input of replica switches **530**. A first output of replica switches **530** provides the signal LS_sw_sense **536**. A second output of replica switches **530** provides the signal LS_gnd_sense **538**. A third output of replica switches **530** provides the signal HS_sw_sense **540**. A fourth output of replica switches **530** provides the signal HS_vdd_sense **542**.

[0056] Comparator **544** has a first input coupled to the first output of replica switches **530** and receives the signal LS_sw_sense **536**. A second input of comparator **544** is coupled to the second output of replica switches **530** and receives the signal LS_gnd_sense **538**. The output of comparator **544** is coupled to the second input of buck control logic circuit **508**, and provides the signal Valley_comp **504**. Comparator **546** has a first input coupled to the third output of replica switches **530** and receives the signal HS_sw_sense **540**. A second input of comparator **546** is coupled to the fourth output of replica switches **530** and receives the signal HS_vdd_sense **542**. The output of comparator **546** is coupled to the third input of buck control logic circuit **508**, and provides the signal Peak_comp **504**.

[0057] The peak current through HS switch **120** and the valley current through LS switch **128** are measured while the power converter is operating in PWM mode, prior to entering PFM mode. The information is provided to Buck control logic circuit **508**, which produces a digital output Sel_i_ripple **432**. The signal Sel_i_ripple **432** is provided as an input to DAC **434**. The value of Sel_i_ripple **434** is ramped down until the inductor current flowing through the HS switch **120** that is measured by HS replica switches **534** and comparator **546** equals the reference level I_peak **210**. The output of DAC **434** is provided as an input to V2I circuit **524** and to PFM timer circuit **525**.

[0058] When operating in the PFM mode and a Start_pfm_pulse occurs, the inductor current ramps up until it reaches the reference level I_peak **210**, which is defined by DAC **434**, and causes the Peak_comp signal **506** at the output of comparator **546** to be asserted. In response to the inductor current reaching the level of I_peak **210**, the inductor current begins decreasing. In response to the inductor current reaching a zero current level, comparator **544** produces a signal at its output, Valley_Comp **504**, which causes buck control logic circuit **508** to produce a Buck_hiz signal **520** to be asserted. Asserting Buck_hiz **520** causes the outputs of driver circuit **184** to be placed into a high impedance state, and no current will flow through HS switch **120** and LS switch **128**. A subsequent start_pfm pulse **426** will trigger, and the PFM cycle will repeat.

[0059] The input to DAC **434** determines the magnitude of I_Ripple **528**, which controls the reference voltage level for comparator **546**. The output of DAC **434** is used in the PFM timer to define the reference level for the timer comparator in PFM timer circuit **525**, which affects the frequency of the PFM pulse. However, in at least one example, a separate DAC can be used for the PFM timer circuit reference level. Some advantages that buck power converter **500** may provide are lower ripple on the output current and voltage signals, and better stability and cleaner transitions switching between PWM mode and PFM mode.

[0060] In this description, “terminal,” “node,” “interconnection,” “lead” and “pin” are used interchangeably. Unless specifically stated to the contrary, these terms generally mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device, or other electronics or semiconductor component.

[0061] In this description, “ground” includes a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground and/or any other form of ground connection applicable to, or suitable for, the teachings of this description.

[0062] In this description, the term “couple” may cover connections, communications or signal

paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action, then: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, so device B is controlled by device A via the control signal generated by device A.

[0063] In this description, even if operations are described in a particular order, some operations may be optional, and the operations are not necessarily required to be performed in that particular order to achieve specified results. In some examples, multitasking and parallel processing may be advantageous. Moreover, a separation of various system components in the embodiments described above does not necessarily require such separation in all embodiments.

[0064] Modifications are possible in the described embodiments, and other embodiments are possible, within the scope of the claims.

Claims

1. A circuit comprising: a comparator having first and second comparator inputs and a comparator output, wherein the second comparator input is coupled to a ripple reference voltage terminal; a switch coupled between the first comparator input and a ground terminal, and having a switch control terminal coupled to the comparator output; a capacitor coupled between the first comparator input and the ground terminal; a first current source coupled between a supply terminal and the first comparator input, and having a first current control terminal; and a second current source coupled between a current output terminal and the ground terminal, and having a second current control terminal.
2. The circuit of claim 1, further comprising a digital-to-analog converter (DAC) having a DAC input and a DAC output, wherein the DAC input is coupled to a ripple control signal terminal, and the DAC output is coupled to the second comparator input and the second current control terminal.
3. The circuit of claim 2, further comprising a control logic circuit having first, second and third control logic inputs and a control logic output, wherein the first control logic input is coupled to a clock source, the second control logic input is coupled to the comparator output, and the control logic output is coupled to the ripple control signal terminal.
4. The circuit of claim 3, wherein the DAC input receives a digital ripple control signal from the ripple control signal terminal, and the DAC output provides an analog conversion of the digital ripple control signal.
5. The circuit of claim 4, wherein the digital ripple control signal includes six bits.
6. The circuit of claim 4, further comprising an amplifier having first and second amplifier inputs and an amplifier output, wherein the first amplifier input is coupled to a voltage regulator feedback terminal, the second amplifier input is coupled to a voltage regulator reference terminal, and the amplifier output is coupled to the first current control terminal.
7. The circuit of claim 6, wherein the capacitor is a first capacitor and the circuit is further comprising a resistor and a second capacitor coupled in series between the amplifier output and the ground terminal.
8. A control circuit comprising: a clock circuit having a clock output; a logic circuit having first, second, third and fourth logic circuit inputs and first, second and third logic circuit outputs, wherein the first logic circuit input is coupled to the clock output; a digital-to-analog converter (DAC) having a DAC input and a DAC output, wherein the DAC input is coupled to the first logic circuit output; a voltage-to-current converter (V2I) having first and second V2I inputs and first, second and third V2I outputs, wherein the first V2I input is coupled to the DAC output; a timer circuit having first, second, and third timer outputs, wherein the first timer output is coupled to the first V2I input, the second timer output is coupled to the second V2I input, and the third timer

output is coupled to the third logic circuit input; and an amplifier having first and second amplifier inputs and an amplifier output, wherein the first amplifier input is coupled to a voltage feedback terminal, the second amplifier input is coupled to a voltage reference terminal, and the amplifier output is coupled to the second V2I input.

9. The control circuit of claim 8, further comprising: a replica circuit having first, second, third, fourth, fifth, sixth, sixth and seventh replica circuit inputs and first, second, third and fourth replica circuit outputs, wherein the first replica circuit input is coupled to the first V2I output, the second replica circuit input is coupled to the second V2I output, the third replica circuit input is coupled to the first V2I output, the fourth replica circuit input is coupled to the third V2I output, and the fifth replica circuit input is coupled to a ground terminal.

10. The control circuit of claim 9, further comprising: a first comparator having first and second comparator inputs and a first comparator output, wherein the first comparator input is coupled to the first replica circuit output, the second comparator input is coupled to the second replica circuit output, and the first comparator output is coupled to the second logic circuit input; and a second comparator having third and fourth comparator inputs and a second comparator output, wherein the third comparator input is coupled to the third replica circuit output, the fourth comparator input is coupled to the fourth replica circuit output, and the second comparator output is coupled to the fourth logic circuit input.

11. The control circuit of claim 9, further comprising a ramp circuit having first and second ramp terminals, wherein the first ramp terminal is coupled to the first replica circuit input, and the second ramp terminal is coupled to the second replica circuit input.

12. The control circuit of claim 10, further comprising a driver circuit having first and second driver circuit inputs and first and second driver circuit outputs, wherein the first driver circuit input is coupled to the second logic circuit output, and the second driver circuit input is coupled to the third logic circuit output.

13. The control circuit of claim 12, further comprising: a high side transistor coupled between an input voltage terminal and a switching terminal and having a high side control terminal, wherein the high side control terminal is coupled to the first driver circuit output; and a low side transistor coupled between the switching terminal and a ground terminal and having a low side control terminal, wherein the low side control terminal is coupled to the second driver circuit output; wherein the sixth replica circuit input is coupled to the switching terminal, and the seventh replica circuit input is coupled to the input voltage terminal.

14. The control circuit of claim 13, further comprising: an inductor coupled between the switching terminal and an output voltage terminal; and a capacitor coupled between the output voltage terminal and the ground terminal.

15. The control circuit of claim 13, wherein the first comparator output is asserted when a current through the switching terminal drops to zero.

16. The control circuit of claim 15, wherein the second comparator output is asserted when the current through the switching terminal reaches a peak value.

17. The control circuit of claim 8, further comprising a resistor and a capacitor coupled in series between the amplifier output and a ground terminal.

18. The control circuit of claim 12, wherein the driver circuit goes into a high impedance responsive to a signal at the second driver circuit input.

19. The control circuit of claim 8, wherein a signal at the first logic circuit output is a digital ripple control signal.

20. The control circuit of claim 19, wherein the digital ripple control signal includes six bits.
